

## UTTAM C. PAWAR

**Phone:** +1 503.929.6681

**Email:** [uttampc@gmail.com](mailto:uttampc@gmail.com)

**Github Profile:** <https://github.com/uttampawar>

**3858, NW Loy Ct.**

Portland, OR 97229

### PROFILE

Strategic systems and performance engineer with many years of experience optimizing runtimes, databases and compiler toolchains for enterprise-class platforms. Proven record of engineering high-performance software for Intel, AMD and IBM, contributing to open-source projects, and delivering deep system-level enhancements. Specialist in C/C++, Linux internals, container performance and x86 architecture.

### TECHNICAL SKILLS

**Languages:** C, C++, Golang, JavaScript, Assembly (x86-64)

**Frameworks:** Node.js, React

**Systems:** Linux, Mac OS X, Unix, Glibc, Kernel internals

**Tools:** Gdb, Perf and vTune profiler, flame graph, LLVM-BOLT, ISA-L

**Open Source:** MongoDB, Node.js, Golang, PHP, Linux kernel, V8 JS runtime, Brotli compression

**Published performance blueprint:**

<https://www.intel.com/content/dam/develop/external/us/en/documents/runtimeperformanceoptimizationblueprint-largecodepages-q1update.pdf>

### EMPLOYMENT HISTORY

**Intel Corporation**, Hillsboro, Oregon Feb 2012-to date **Cloud Runtime Software Engineer**

At Intel I've worked on a diverse set of product performance (single and multicore scaling) issues which involved performance data analysis, identification of CPU micro-architecture (CPU frontend, branch predictor efficiency, cache lines and backend stalls) and un-core bottlenecks with the help of tools such as, "Perf" and "Emon" (CPU core event monitor).

- 1) Achieved 750% throughput improvement in "Brotli" compression via low-level CPU profiling and runtime optimization.
- 2) Delivered 5x MongoDB query performance boost on Xeon through core execution path and cycle improvements.
- 3) Led a small team of 4, and directed performance enhancements across top 10 datacenter containers (Node.js, Nginx, Redis, PHP+Wordpress, etc.)
- 4) Tuned Golang and PHP runtimes to leverage x86 micro-architecture advantages.
- 5) Developed optimized JavaScript NPM module with binding to ISAL (<https://github.com/intel/isa-l>) library exposing accelerated encryption and compression support in the Xeon processor.
- 6) Refined Android core runtime to reduce power consumption and latency on Intel client platform.
- 7) Actively maintaining many open-source projects, viz., NodeJS @ <https://github.com/nodejs>, a repository for IA users, offering tools and a reference implementation for optimizing dynamic runtimes called IODLR @ <https://github.com/intel/iodlr> and a datacenter workload for NodeJS project at <https://github.com/Node-DC>. This was developed and contributed to the upstream NodeJS project.
- 8) Interacted with many Intel's end customers directly diagnosing real-time performance issues.

**Advanced Micro Devices (AMD),** Portland, Oregon 2006–2011 **Member of Technical Staff**

- 1) Optimized compiler code generation and loop vectorization in PGI/Open64 for AMD Opteron based platform.
- 2) Boosted SPEC CPU 2000/2006 benchmark performance significantly through profiling and targeted tuning.
- 3) Honored with AMD Spotlight and Spirit of Success awards for outstanding technical contributions.

**International Business Machines (IBM),** Beaverton, Oregon 2000–2006 **Software Engineer**

- 1) Improved GCC tool chain and Glibc performance for PowerPC architecture; contributed upstream patches.
- 2) Ported libnuma library for SLES9 OS on Power platform and maintained NUMA support for the same.
- 3) Boosted SPEC CPU 2000 benchmark performance on PowerPC platform.
- 4) Submitted a disclosure for 'source code browser/analyzer which improved the time it took to study new project source code written in C or C++ programming language.
- 5) Enhanced Linux driver subsystem and scheduler code for IBM hardware compatibility.

**MBNA AMERICA,** Wilmington, Delaware 1998–2000 **Software Programmer Consultant**

- 1) Created complete architecture, detail design documents and developed Auto Debt Management Automation system in C++ generating revenue over \$2M per year.
- 2) Contributed to development of C++ based backend services for "Settlement Automation", "Loss Prevention Forms Automation" and "Fraud Detection System" in credit card processing resulting in improved customer service satisfaction. Above projects involved requirement gathering, creation of detailed design documents and development.

**AT&T,** Bell Labs, New Jersey 1997–1998 **System Software Programmer**

- 1) Contributed development and testing of C++ client-server application which monitored inter-toll network using Tuxedo as middleware software ensuring the system that managed the switching/regulation of calls was operating properly.
- 2) Supported the provisioning and maintenance of echo Cancelers system, a C++ application.

**ChartLync LLC. | Founder |** Portland, OR | 2015-to present

- 1) Created this company developing unique technology to address complexity in sharing of patient "Health Records".
- 2) Designmobile/web platform for secure patient record sharing.
- 3) Led all development phase from user research to deployment (production release) on iOS and Android platforms.

**EDUCATION HISTORY**

Graduate Coursework in Computer Science, Oregon Graduate Institute, PSU. Portland, OR, USA

**BSc.** Computer Engineering, University of Mumbai, India

**Diploma in Computer Technology** University of Mumbai, India